PROBLEM DA3

Stirrer in a Tar Tank

Original statement · DA3.pdf

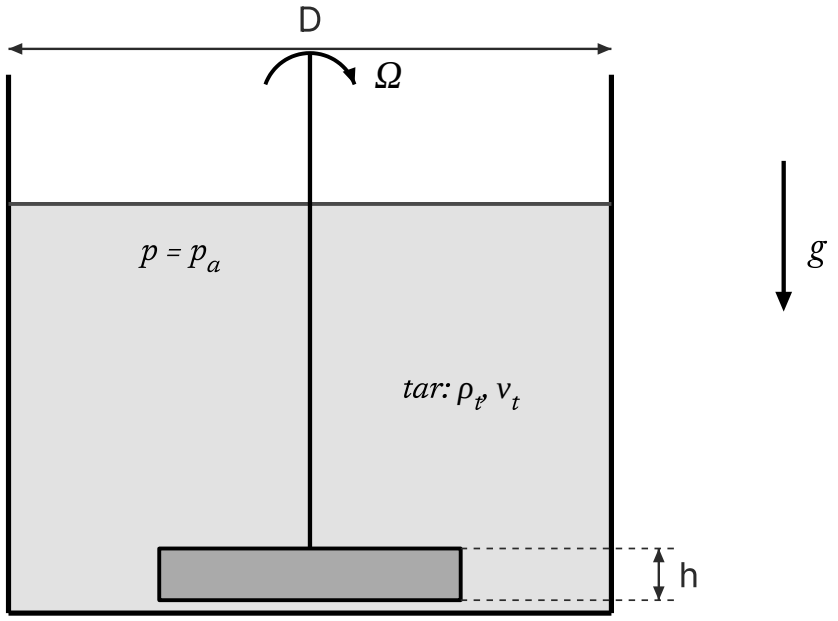
Π theorem | Froude and Reynolds | Change of fluid

OFFICIAL PROBLEM STATEMENT

A large container of diameter $D = 10$ m contains a volume $V = 1000$ m^3 of tar ($\nu_t = 10^{-3}$ m^2/s, $\rho_t = 2 \times 10^3$ kg/m^3). To avoid solidification, the tar is continuously moved with a 3-blade stirrer of height $h = 1$ m placed at its bottom, which rotates with angular velocity $\Omega = 1$ s^{-1}, as shown in the figure. Design a small experiment with a geometrically similar model to determine the power W needed to operate the stirrer. In particular,

- Use the Π theorem to reduce the parametric dependence of W.
- By using water in the experiment as working fluid, obtain the values of D_e, h_e, V_e y Ω_e required to guarantee that the experiment is physically similar to the prototype.
- Obtain the ratio W/W_e between the power measured in the experiment W_e and that required to move the stirrer of the real prototype.

The sheet writes "y", Spanish for "and", in the second question.



volume V of tar; 3-blade stirrer of height h at the bottom

Statement figure (redrawn from DA3.pdf, section view): a container of diameter D holding a volume V of tar, free surface at p_a, with a 3-blade stirrer of height h at its bottom rotating at Ω; gravity g downward

Source: DA3.pdf (page 1, statement and handwritten key). Notes.pdf, Chapter 6: Π theorem, Eq. 6.25; physical similarity, Eqs. 6.57-6.59.

Worked solution

Hypotheses, Data and Dimensional Table

- The tar has a free surface at $p = p_a$ that the stirrer deforms, so gravity enters through that boundary condition (Notes.pdf, p. 66); the viscosity of tar is large, so viscosity enters too.
- Geometry: the tank diameter D, the stirrer height h and the volume V (which fixes the depth of tar); the blade shape is fixed by geometric similarity.
- Water in the model: $\nu_w = 10^{-6}$ m^2/s, $\rho_w = 10^3$ kg/m^3, so $\mu_w = 10^{-3}$ kg/(m s). Tar: $\mu_t = \rho_t \nu_t = 2 \times 10^3 \times 10^{-3} = 2$ kg/(m s).

Parameter	Dimensions	Prototype
W (unknown)	ML^2T^{-3}	?

Parameter	Dimensions	Prototype
Ω	T^{-1}	1 s^{-1}
g	LT^{-2}	9.81 $\mathrm{m/s}^2$
h	L	1 m
D	L	10 m
V	L^3	1000 m^3
ρ	ML^{-3}	2×10^3 $\mathrm{kg/m}^3$
μ	$ML^{-1}T^{-1}$	2 kg/(m s)

Application of the Π Theorem

$$W = W(\Omega, g, h, D, V, \rho, \mu)$$

$n = 7$. The key takes (μ, Ω, h) as the independent set, natural for a very viscous liquid (Notes.pdf, p. 68). Rows Ω, h, μ; columns L, M, T:

$$\begin{pmatrix} 0 & 0 & -1 \\ 1 & 0 & 0 \\ -1 & 1 & -1 \end{pmatrix}, \qquad \det = (-1)\cdot(1\cdot 1 - 0\cdot(-1)) = -1 \neq 0$$

(expanding along the first row), so $k = 3$ and $n - k = 4$ groups remain.

Power. $[W] = [\mu]^a[\Omega]^b[h]^c$: $M:\ 1 = a$; $T:\ -3 = -a - b$; $L:\ 2 = -a + c$; so $a = 1$, $b = 2$, $c = 3$: $\Pi_0 = W/(\mu\Omega^2 h^3)$.

Gravity. $[g] = [\Omega]^b[h]^c$: $T:\ -2 = -b$; $L:\ 1 = c$: the group $g/(\Omega^2 h)$, an inverse Froude number squared.

Geometry. D/h and V/h^3.

Density. $[\rho] = [\mu]^a[\Omega]^b[h]^c$: $M:\ 1 = a$; $T:\ 0 = -a - b$, so $b = -1$; $L:\ -3 = -a + c$, so $c = -2$: $\rho/(\mu\Omega^{-1}h^{-2}) = \rho\Omega h^2/\mu$, a Reynolds number.

$$\frac{W}{\mu\Omega^2 h^3} = f\left(\frac{g}{\Omega^2 h}, \frac{D}{h}, \frac{V}{h^3}, \frac{\rho\Omega h^2}{\mu}\right)$$

Physical Similarity with Water

All four groups must be equal in model (e, water) and prototype (t, tar), with the same g:

$$\frac{g}{\Omega^2 h} = \frac{g}{\Omega_e^2 h_e} \implies \frac{h_e}{h} = \left(\frac{\Omega}{\Omega_e}\right)^2$$

$$\frac{\Omega h^2}{\nu_t} = \frac{\Omega_e h_e^2}{\nu_w} \implies \left(\frac{h_e}{h}\right)^2 = \frac{\Omega}{\Omega_e}\,\frac{\nu_w}{\nu_t}$$

$$\frac{D}{h} = \frac{D_e}{h_e}, \qquad \frac{V}{h^3} = \frac{V_e}{h_e^3}$$

The first two fix the scale and the speed. From the first, $\Omega/\Omega_e = (h_e/h)^{1/2}$; into the second:

$$\left(\frac{h_e}{h}\right)^2 = \left(\frac{h_e}{h}\right)^{1/2}\frac{\nu_w}{\nu_t} \implies \left(\frac{h_e}{h}\right)^{3/2} = \frac{\nu_w}{\nu_t} \implies \frac{h_e}{h} = \left(\frac{\nu_t}{\nu_w}\right)^{-2/3} = \left(10^3\right)^{-2/3} = 10^{-2}$$

$$h_e = 10^{-2}\times 1 = 1\ \text{cm}, \qquad \frac{\Omega_e}{\Omega} = \left(\frac{h}{h_e}\right)^{1/2} = \left(\frac{\nu_t}{\nu_w}\right)^{1/3} = 10 \implies \Omega_e = 10\ \text{s}^{-1}$$

$$D_e = \frac{h_e}{h}\,D = 10^{-2}\times 10 = 10\ \text{cm}, \qquad V_e = \left(\frac{h_e}{h}\right)^3 V = 10^{-6}\times 1000 = 10^{-3}\ \text{m}^3 = 1\ \text{litre}$$

This is the way out of the Reynolds-Froude incompatibility of DA1: the model fluid must satisfy $\nu_w/\nu_t = (h_e/h)^{3/2}$, and water does it for a 1:100 model of a tank of tar.

Check of both groups: prototype $\Omega h^2/\nu_t = 1\times 1/10^{-3} = 10^3$, model $\Omega_e h_e^2/\nu_w = 10\times 10^{-4}/10^{-6} = 10^3$; prototype $g/(\Omega^2 h) = 9.81$, model $9.81/(100\times 0.01) = 9.81$.

Power Ratio

With all groups equal, Π_0 is equal:

$$\frac{W}{\mu_t \Omega^2 h^3} = \frac{W_e}{\mu_w \Omega_e^2 h_e^3} \implies \frac{W}{W_e} = \frac{\mu_t}{\mu_w}\left(\frac{\Omega}{\Omega_e}\right)^2\left(\frac{h}{h_e}\right)^3 = \frac{\rho_t}{\rho_w}\,\frac{\nu_t}{\nu_w}\left(\frac{\Omega}{\Omega_e}\right)^2\left(\frac{h}{h_e}\right)^3$$

$$\frac{W}{W_e} = 2\times 10^3 \times \left(10^{-1}\right)^2 \times \left(10^2\right)^3 = 2\times 10^3 \times 10^{-2} \times 10^6 = 2\times 10^7$$

Answers and Checks

- Reduced law: $W/(\mu\Omega^2 h^3) = f\left(g/(\Omega^2 h),\ D/h,\ V/h^3,\ \rho\Omega h^2/\mu\right)$.
- Model in water: $h_e = 1$ cm, $D_e = 10$ cm, $V_e = 1$ litre, $\Omega_e = 10\ \text{s}^{-1}$.
- Power ratio: $W/W_e = 2\times 10^7$.
- Dimensions: $\mu\Omega^2 h^3$ is $ML^{-1}T^{-1}\cdot T^{-2}\cdot L^3 = ML^2T^{-3}$, a power.
- Consistency: the same ratio follows from $\rho\Omega^3 h^5$ as power scale (the other choice of the independent set): $(\rho_t/\rho_w)(\Omega/\Omega_e)^3(h/h_e)^5 = 2\times 10^{-3}\times 10^{10} = 2\times 10^7$.
- Check against the official key: DA3.pdf gives $h_e = 1$ cm, $\Omega_e = 10\ \text{s}^{-1}$, $D_e = 10$ cm, $V_e = 1$ litre and $W/W_e = 2\times 10^7$.